DT 111 · TOPIC 6

Artificial Intelligence Basics

Search, reasoning, neural networks and the AI toolbox

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain AI as a broad field rather than one algorithm.
- Distinguish AI, Machine Learning, Deep Learning and Generative AI.
- Describe search problems using states, actions and goals.
- Compare BFS, DFS and heuristic search conceptually.
- Explain rule-based reasoning and a simple artificial neuron.

AI is a toolbox

Search and planning

Explore possible states to reach a goal.

Logic and reasoning

Apply explicit facts and rules to derive conclusions.

Machine Learning

Learn patterns from examples.

Neural networks

Learn complex representations from data.

Generative AI

Generate text, images, audio, code and other content.

AI, ML, Deep Learning and Generative AI

Artificial Intelligence

Broadest field: systems performing tasks associated with intelligent behaviour.

Machine Learning

AI approach based on learning from data.

Deep Learning

Machine Learning using multi-layer neural networks.

Generative AI

Models designed to create new content; many modern systems use deep learning.

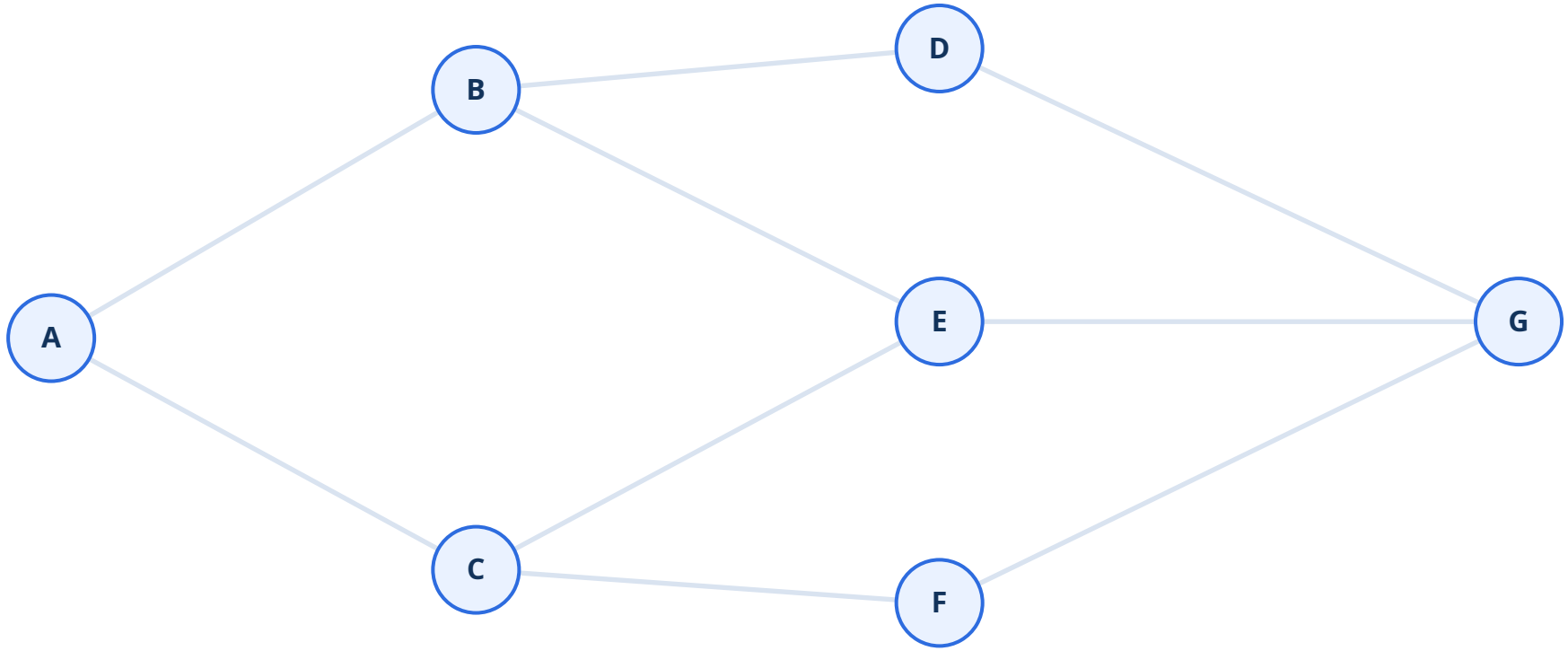


Search as problem solving

Many AI problems can be represented as a starting state, possible actions and a goal state.

A small state space

Start at A and search for a path to G. Different algorithms explore the same graph in different orders.



Breadth-First Search and Depth-First Search

BFS

- Explores level by level
- Uses a queue
- Finds a fewest-edge path in an unweighted graph
- Can require substantial memory

DFS

- Explores one branch deeply first
- Uses a stack or recursion
- Memory can be smaller
- May reach a long or unhelpful branch first

A* search adds a heuristic

Path cost g(n)

Cost already spent to reach the current state.

Heuristic h(n)

Estimate of remaining cost to the goal.

A* priority

f(n) = g(n) + h(n). Promising states are explored first.

A heuristic is an estimate

Useful heuristic

- Provides guidance toward the goal
- Cheap enough to compute
- Reflects meaningful problem structure

Poor heuristic

- Gives little useful guidance
- Can waste search effort
- May behave almost like uninformed search

Logic and reasoning

Symbolic AI represents knowledge explicitly as facts, rules and inference procedures.

Example: rule-based reasoning

1

Fact

Soil is dry

2

Fact

Rain is not expected

3

Rule

IF dry AND no rain → irrigation needed

4

Inference

Irrigation is needed

Strengths and limits of symbolic rules

Strength

Reasoning can be transparent and easy to inspect.

Strength

Domain knowledge can be encoded explicitly.

Limit

Rules may become difficult to maintain at scale.

Limit

Hard to write complete rules for perception or natural language.



Neural networks

Neural networks learn weighted combinations of inputs and can model complex non-linear relationships.

A simple artificial neuron

1

Inputs

x1, x2, x3

2

Weights

Each input has influence

3

Sum + bias

Combine weighted inputs

4

Activation

Transform the combined signal

5

Output

Pass information forward

Neural-network layers

Input layer

Receives features or encoded information.

Hidden layers

Transform representations through learned weights.

Output layer

Produces a prediction, score or representation.

Traditional ML and deep learning

Traditional ML

- Often relies on designed features
- Can work very well on structured/tabular data
- Usually smaller models

Deep learning

- Learns complex representations automatically
- Especially effective for images, speech and language
- Often requires more data and computation



From neural networks to generative AI

Large neural models can learn patterns in text, images, audio and code and use those patterns to generate new content.

Choosing an AI method

The correct method depends on the problem, data, constraints and risks.

Route planning

Search / planning algorithms.

Expert rules

Logic and symbolic reasoning.

Tabular prediction

Supervised Machine Learning.

Image recognition

Often deep neural networks.

Content generation

Generative models.

AI capability and human judgement

AI can help

- Search large spaces
- Recognise patterns
- Automate repeated decisions
- Generate candidate content

Humans remain responsible for

- Defining goals and values
- Checking outputs and context
- Managing risk and accountability
- Deciding how systems should be used

Topic 6 key ideas

AI contains multiple approaches: search, logic, Machine Learning and neural networks.

BFS, DFS and A* differ mainly in how they choose which state to explore next.

Rules make knowledge explicit; learning methods derive patterns from data.

Deep Learning is a subset of Machine Learning, not a synonym for all AI.

Method choice should follow the problem, not current technology trends.